

**ELECTRICAL INTERFACE CONTROL DOCUMENT**  
**FOR**  
**ELECTRONIC FLIGHT INSTRUMENT SYSTEM**  
**ELECTRONIC FLIGHT INSTRUMENT**  
**ASTRONAUTICS P/N 195000-11**  
**CONTROL PANEL**  
**ASTRONAUTICS P/N 195010-11**

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|  <b>Astronautics Corporation of America</b><br><b>Mailing Address</b><br>Post Office Box 523<br>Milwaukee, Wisconsin 53201<br><b>Corporate Headquarters</b><br>4115 North Teutonia Ave.<br>Milwaukee, Wisconsin 53209 | <b>CAGE Code</b><br>10138     |
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**List of Acronyms**


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|       |                                          |
|-------|------------------------------------------|
| AC    | Aircraft Configuration                   |
| AC    | Alternating Current                      |
| A/C   | Aircraft                                 |
| ADI   | Attitude Director Indicator              |
| ALT   | Altimeter                                |
| ALT   | Alternate                                |
| AMLCD | Active Matrix Liquid Crystal Display     |
| ARINC | Aeronautical Radio Incorporated          |
| ATP   | Acceptance Test Procedure                |
| ATT   | Attitude                                 |
| CADI  | Copilot Attitude Director Indicator      |
| CAP   | Capture                                  |
| CHAS  | Chassis                                  |
| CHSI  | Copilot Horizontal Situation Indicator   |
| CIWS  | Central Instrument Warning System        |
| CMD   | Command                                  |
| COMB  | Combined ADI/HSI Display Mode            |
| CP    | Control Panel                            |
| CRS   | Course                                   |
| CTR   | Center                                   |
| CWX   | Color Weather Radar                      |
| DC    | Direct Current                           |
| d/c   | Don't Care                               |
| DEV   | Deviation                                |
| DH    | Decision Height                          |
| DIAG  | Diagnostic                               |
| DME   | Distance Measuring Equipment             |
| EFI   | Electronic Flight Instrument             |
| EFIS  | Electronic Flight Instrument System      |
| EGPWS | Enhanced Ground Proximity Warning System |
| ETI   | Elapsed Time Indicator                   |
| FD    | Flight Director                          |
| FMS   | Flight Management System                 |
| GND   | Ground                                   |
| GPWS  | Ground Proximity Warning System          |
| G/S   | Glideslope                               |
| GS    | Glideslope                               |
| H     | High (reference voltage)                 |
| HDG   | Heading                                  |
| HS    | High Speed                               |
| HSI   | Horizontal Situation Indicator           |
| Hz    | Hertz                                    |
| ICD   | Interface Control Document               |
| ILS   | Instrument Landing System                |
| IM    | Inner Marker Beacon                      |
| IN    | Input                                    |
| INS   | Inertial Navigation System               |
| IRU   | Inertial Reference Unit                  |
| L     | Low (reference voltage)                  |
| LS    | Low Speed                                |
| LMP   | Lamp                                     |
| LNAV  | Lateral Navigation Mode                  |
| LOC   | Localizer                                |
| LRU   | Line Replaceable Unit                    |

|       |                                              |
|-------|----------------------------------------------|
| MAG   | Magnetic                                     |
| MAINT | Maintenance Mode                             |
| MAP   | Map Overlay                                  |
| MSTR  | Master                                       |
| NAV   | Navigation                                   |
| NM    | Nautical Mile                                |
| OUT   | Output                                       |
| PADI  | Pilot Attitude Director Indicator            |
| PHSI  | Pilot Horizontal Situation Indicator         |
| PMS   | Performance Management System                |
| P/N   | Part Number                                  |
| RALT  | Radio Altimeter                              |
| REF   | Reference                                    |
| RTE   | Route                                        |
| SEL   | Selected                                     |
| SIG   | Signal                                       |
| SPD   | Speed                                        |
| SRC   | Source                                       |
| SRD   | Software Requirements Document               |
| SYNC  | Synchronous                                  |
| TAWS  | Terrain Avoidance and Warning System         |
| TCAS  | Traffic Alert and Collision Avoidance System |
| T/F   | To/From                                      |
| TE    | Test Equipment                               |
| TERR  | Terrain                                      |
| TFC   | Traffic                                      |
| TSP   | Twisted Shielded Pair                        |
| TST   | Twisted Shielded Triple                      |
| TTL   | Tuned to Localizer                           |
| V     | Volt                                         |
| VAL   | Valid                                        |
| VERT  | Vertical                                     |
| VOR   | Very High Frequency Omni-range Station       |
| WOW   | Weight On Wheels                             |
| WYPT  | Waypoint                                     |
| XLINK | Crosslink                                    |

# 1 SCOPE

This document identifies the electrical interfaces utilized by the Electronic Flight Instrument System (EFIS) developed for the Transaero 747 Program. A ship-set consists of four Electronic Flight Instruments (Astronautics P/N 195000-11) and two Control Panels (Astronautics P/N 195010-11).

## 1.1 Electronic Flight Instrument and Control Panel Description

The Electronic Flight Instrument for the Transaero Program is a 5ATI Active Matrix Liquid Crystal Display (AMLCD) which contains aircraft interface circuitry, graphics generation, associated display and power circuitry and a Control Panel for user interface.

The Transaero EFIS consists of four identical EFIs and two CPs. One Attitude Direction Indicator (ADI), one Horizontal Situation Indicator (HSI) and one Control Panel will be located on each side of the cockpit. The EFIs are 5ATI units, which contain interfaces to process discrete, analog, RS422 and ARINC 429, ARINC 453 and ARINC 568 aircraft inputs to generate the primary and navigational flight displays. The CP contains discrete rotary knobs and pushbuttons that connect directly to the EFIs.

The EFI uses a 256 shade of gray 4"x4" AMLCD of which 64 shades (for each color) are used to provide a high resolution, high brightness color display presentation. The EFI bezel has 2 pushbuttons for brightness control and an inclinometer. The CP has 9 pushbuttons and 2 rotary knobs for control of EFI functionality such as Range and Mode selections.

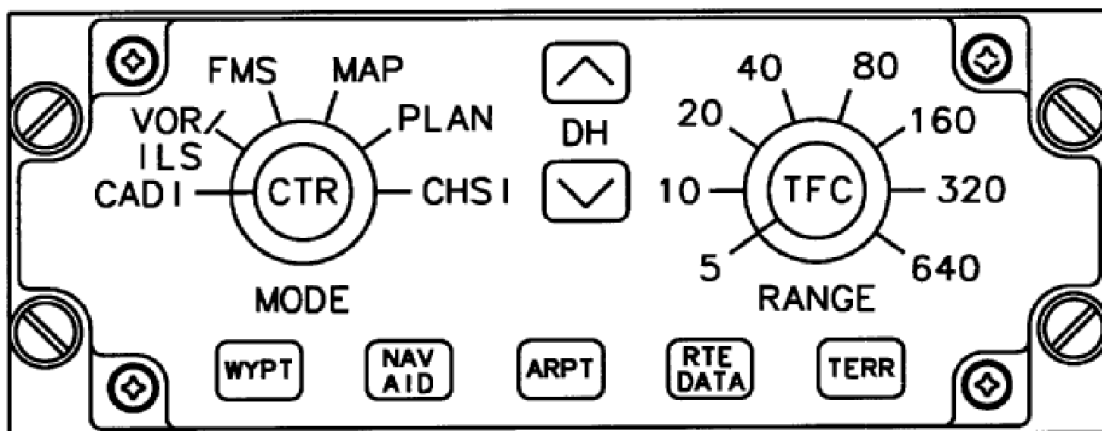


Figure 1 Control Panel (CP)

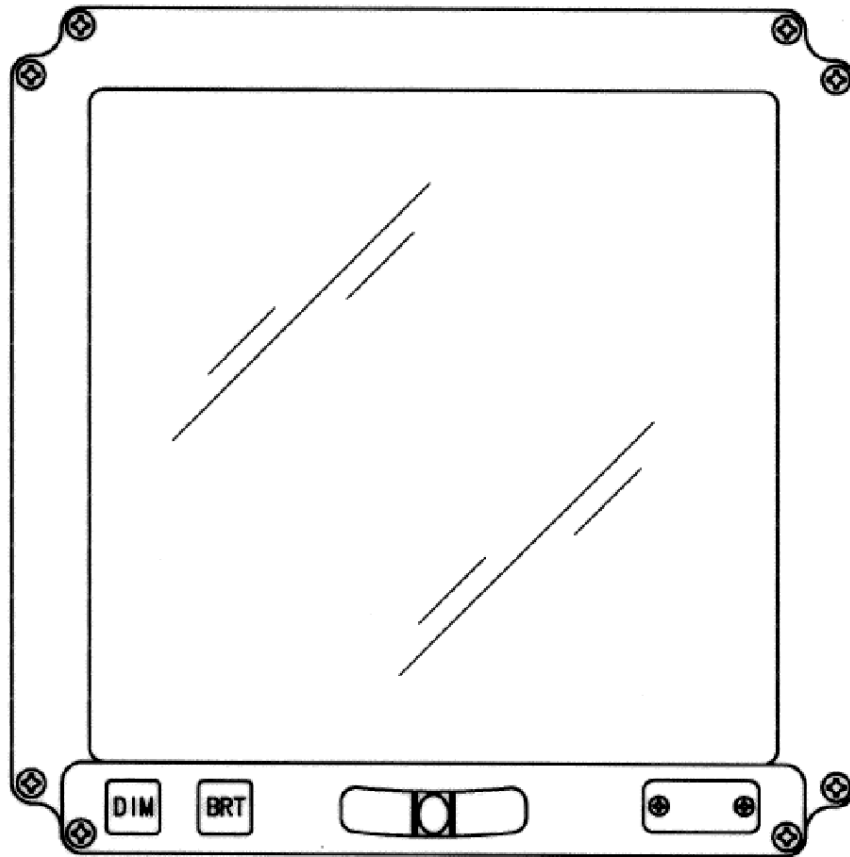


Figure 2: ELECTRONIC FLIGHT INSTRUMENT (EFI)

## 2 APPLICABLE DOCUMENTS

The following documents, of the exact issue shown or of the latest issue in effect at the time of this document, form a part of this document to the extent specified herein. In the event of a conflict between the referenced documents and the contents of this document, the contents of this document shall take precedence to the extent specified.

|                |                                                            |
|----------------|------------------------------------------------------------|
| SRD195544-xxxx | Software Requirements Document (xxxx = latest version)     |
| ATP09273x      | Acceptance Test Procedure for the EFI (x = latest version) |
| ATP09276x      | Acceptance Test Procedure for the CP (x = latest version)  |
| ARINC: 407     | Synchro Systems Manual                                     |
| ARINC: 408     | Air Transport Indicator Cases and Mounting                 |
| ARINC: 429     | Mark 33 Digital Information Transfer System                |
| ARINC: 453     | Airborne Weather Radar Digital Data                        |
| ARINC: 561     | Air Transport Inertial Navigation System                   |
| ARINC: 735     | Traffic Alert and Collision Avoidance System               |

### 3 EFI SUBSYSTEM INTERFACES

#### 3.1 Electrical Interface Pinouts

##### 3.1.1 EFI and CP Connectors

The EFI electrical interfaces are through three D38999 Series II style connectors. The J1 connector has 55 filtered pins and has the master key way in the normal position. The J2 connector has 79 filtered pins and has the master key way in the normal position. The J3 connector has 79 filtered pins and has the master key way in the A-position. The CP electrical interfaces are through one D38999 Series II style connector. The J1 connector has 41 pins and has the master key way in the normal position

The mechanical interfaces to the connectors are as follows:

EFI J1 - equivalent to MS27508E22F55P, mates with MS27473E22F55S or equivalent.

EFI J2 - equivalent to MS27508E20A35P, mates with MS27473E20F35S or equivalent.

EFI J3 - equivalent to MS27508E20A35PA, mates with MS27473E20F35SA or equivalent.

CP J1 - equivalent to MS27508E20F41P, mates with MS27473E20F41S or equivalent.

##### 3.1.2 Electronic Flight Instrument Pinout

###### 3.1.2.1 EFI J1 Pinout

Table 1: EFI J1 Pinout

| EFI J1 Pinout |             |                      |                          |
|---------------|-------------|----------------------|--------------------------|
| J1            | SIGNAL NAME | SIGNAL TYPE          | PARAGRAPH                |
| A             |             | 28V/O DISCRETE INPUT | <a href="#">3.2.7.45</a> |
| B             |             | 28V/O DISCRETE INPUT | <a href="#">3.2.7.70</a> |
| C             | 28VDC       | 28VDC PRIMARY POWER  | <a href="#">3.2.9.1</a>  |
| D             | 28VDC       | 28VDC PRIMARY POWER  | <a href="#">3.2.9.1</a>  |
| E             | 28VDC RTN   | 28VDC PRIMARY RETURN | <a href="#">3.2.9.1</a>  |
| F             | 28VDC RTN   | 28VDC PRIMARY RETURN | <a href="#">3.2.9.1</a>  |
| G             |             | G/O DISCRETE OUTPUT  | <a href="#">3.2.8.26</a> |
| H             |             | G/O DISCRETE OUTPUT  | <a href="#">3.2.8.27</a> |
| J             |             |                      |                          |
| K             | 5VAC        | 5VAC POWER           | <a href="#">3.2.9.2</a>  |
| L             | 5VAC RTN    | 5VAC RETURN          | <a href="#">3.2.9.2</a>  |
| M             | SEL REF H   | REFERENCE            | <a href="#">3.2.6.1</a>  |
| N             |             | REFERENCE            | <a href="#">3.2.6.2</a>  |
| P             |             | REFERENCE            | <a href="#">3.2.6.2</a>  |
| R             | GS H        | ANALOG INPUT         | <a href="#">3.2.11.9</a> |
| S             |             | SYNCHRO INPUT        | <a href="#">3.2.5.5</a>  |
| T             |             | SYNCHRO INPUT        | <a href="#">3.2.5.5</a>  |
| U             |             | SYNCHRO INPUT        | <a href="#">3.2.5.5</a>  |
| V             | CIWS RTN    | GROUND               | <a href="#">3.2.10.3</a> |
| W             |             | G/O DISCRETE INPUT   | <a href="#">3.2.7.71</a> |
| X             | CHAS GND    | CHASSIS GROUND       | <a href="#">3.2.10.1</a> |
| Y             | 28VDC       | 28VDC PRIMARY POWER  | <a href="#">3.2.9.1</a>  |
| Z             | MAG HDG Z   | SYNCHRO INPUT        | <a href="#">3.2.5.3</a>  |
| a             | SEL CRS X   | SYNCHRO INPUT        | <a href="#">3.2.5.2</a>  |
| b             | 28VDC RTN   | 28VDC PRIMARY RETURN | <a href="#">3.2.9.1</a>  |
| c             | T/F H       | ANALOG INPUT         | <a href="#">3.2.11.1</a> |

| EFI J1 Pinout |              |                    |                          |
|---------------|--------------|--------------------|--------------------------|
| J1            | SIGNAL NAME  | SIGNAL TYPE        | PARAGRAPH                |
| d             | T/F L        | ANALOG INPUT       | <a href="#">3.2.11.1</a> |
| e             |              | ANALOG OUTPUT      | <a href="#">3.2.14.1</a> |
| f             |              | ANALOG OUTPUT      | <a href="#">3.2.14.1</a> |
| g             | ROLL CMD H   | ANALOG INPUT       | <a href="#">3.2.11.5</a> |
| h             | SEL REF L    | REFERENCE          | <a href="#">3.2.6.1</a>  |
| i             | SPD DEV H    | ANALOG INPUT       | <a href="#">3.2.11.4</a> |
| j             | SPD DEV L    | ANALOG INPUT       | <a href="#">3.2.11.4</a> |
| k             | GS L         | ANALOG INPUT       | <a href="#">3.2.11.9</a> |
| m             | LOC (CRS) H  | ANALOG INPUT       | <a href="#">3.2.11.2</a> |
| n             | LOC (FINE) H | ANALOG INPUT       | <a href="#">3.2.11.3</a> |
| p             |              | G/O DISCRETE INPUT | <a href="#">3.2.7.72</a> |
| q             |              | G/O DISCRETE INPUT | <a href="#">3.2.7.73</a> |
| r             | SEL HDG X    | SYNCHRO INPUT      | <a href="#">3.2.5.1</a>  |
| s             | MAG HDG X    | SYNCHRO INPUT      | <a href="#">3.2.5.3</a>  |
| t             | SEL CRS Y    | SYNCHRO INPUT      | <a href="#">3.2.5.2</a>  |
| u             | SEL CRS Z    | SYNCHRO INPUT      | <a href="#">3.2.5.2</a>  |
| v             |              | ANALOG OUTPUT      | <a href="#">3.2.14.2</a> |
| w             |              | ANALOG OUTPUT      | <a href="#">3.2.14.2</a> |
| x             | ROLL CMD L   | ANALOG INPUT       | <a href="#">3.2.11.5</a> |
| y             | VERT DEV H   | ANALOG INPUT       | <a href="#">3.2.11.6</a> |
| z             | LOC (CRS) L  | ANALOG INPUT       | <a href="#">3.2.11.2</a> |
| AA            | LOC (FINE) L | ANALOG INPUT       | <a href="#">3.2.11.3</a> |
| BB            |              | G/O DISCRETE INPUT | <a href="#">3.2.7.74</a> |
| CC            | SEL HDG Y    | SYNCHRO INPUT      | <a href="#">3.2.5.1</a>  |
| DD            | MAG HDG Y    | SYNCHRO INPUT      | <a href="#">3.2.5.3</a>  |
| EE            |              |                    |                          |
| FF            | VERT DEV L   | ANALOG INPUT       | <a href="#">3.2.11.6</a> |
| GG            | SIG GND      | SIGNAL GROUND      | <a href="#">3.2.10.2</a> |
| HH            | SEL HDG Z    | SYNCHRO INPUT      | <a href="#">3.2.5.1</a>  |

## 3.1.2.2 EFI J2 Pinout

Table 2: EFI J2 Pinout

| EFI J2 Pinout |              |                       |                          |
|---------------|--------------|-----------------------|--------------------------|
| J2            | SIGNAL NAME  | SIGNAL TYPE           | PARAGRAPH                |
| 1             | RAD ALT H    | ANALOG INPUT          | <a href="#">3.2.11.8</a> |
| 2             | RAD ALT L    | ANALOG INPUT          | <a href="#">3.2.11.8</a> |
| 3             | PITCH CMD H  | ANALOG INPUT          | <a href="#">3.2.11.7</a> |
| 4             | PITCH CMD L  | ANALOG INPUT          | <a href="#">3.2.11.7</a> |
| 5             | EFIS VAL OUT | 28V/O DISCRETE OUTPUT | <a href="#">3.2.8.1</a>  |
| 6             | MSTR LMP     | 28V/O DISCRETE OUTPUT | <a href="#">3.2.8.2</a>  |
| 7             | AC 6         | G/O DISCRETE INPUT    | <a href="#">3.2.7.44</a> |
| 8             | WOW          | G/O DISCRETE INPUT    | <a href="#">3.2.7.1</a>  |
| 9             | IRU ALT      | G/O DISCRETE INPUT    | <a href="#">3.2.7.46</a> |
| 10            | CTR          | G/O DISCRETE INPUT    | <a href="#">3.2.7.2</a>  |
| 11            | VOR Z        | SYNCHRO INPUT         | <a href="#">3.2.5.4</a>  |
| 12            | VOR Y        | SYNCHRO INPUT         | <a href="#">3.2.5.4</a>  |
| 13            | VOR X        | SYNCHRO INPUT         | <a href="#">3.2.5.4</a>  |
| 14            | HDG VAL      | 28V/O DISCRETE INPUT  | <a href="#">3.2.7.3</a>  |
| 15            | GS VAL       | 28V/O DISCRETE INPUT  | <a href="#">3.2.7.47</a> |
| 16            |              | 28V/O DISCRETE INPUT  | <a href="#">3.2.7.48</a> |
| 17            | LOC VAL      | 28V/O DISCRETE INPUT  | <a href="#">3.2.7.4</a>  |
| 18            | FD VAL       | 28V/O DISCRETE INPUT  | <a href="#">3.2.7.5</a>  |
| 19            | SPD DEV VAL  | 28V/O DISCRETE INPUT  | <a href="#">3.2.7.49</a> |
| 20            | RAD ALT VAL  | 28V/O DISCRETE INPUT  | <a href="#">3.2.7.50</a> |
| 21            | ATT LMP      | 28V/O DISCRETE OUTPUT | <a href="#">3.2.8.3</a>  |
| 22            | HDG LMP      | 28V/O DISCRETE OUTPUT | <a href="#">3.2.8.4</a>  |
| 23            |              | SYNCHRO INPUT         | <a href="#">3.2.5.6</a>  |
| 24            |              | SYNCHRO INPUT         | <a href="#">3.2.5.6</a>  |
| 25            |              | SYNCHRO INPUT         | <a href="#">3.2.5.6</a>  |
| 26            |              | SYNCHRO INPUT         | <a href="#">3.2.5.7</a>  |
| 27            |              | SYNCHRO INPUT         | <a href="#">3.2.5.7</a>  |
| 28            |              | SYNCHRO INPUT         | <a href="#">3.2.5.7</a>  |
| 29            | WYPT         | G/O DISCRETE INPUT    | <a href="#">3.2.7.6</a>  |
| 30            | COMB         | G/O DISCRETE INPUT    | <a href="#">3.2.7.51</a> |
| 31            | MAP          | G/O DISCRETE INPUT    | <a href="#">3.2.7.7</a>  |
| 32            | FMS          | G/O DISCRETE INPUT    | <a href="#">3.2.7.8</a>  |
| 33            | PLAN         | G/O DISCRETE INPUT    | <a href="#">3.2.7.9</a>  |
| 34            | TERR         | G/O DISCRETE INPUT    | <a href="#">3.2.7.52</a> |
| 35            | DH UP        | G/O DISCRETE INPUT    | <a href="#">3.2.7.10</a> |
| 36            | DH DOWN      | G/O DISCRETE INPUT    | <a href="#">3.2.7.11</a> |
| 37            | 5NM          | G/O DISCRETE INPUT    | <a href="#">3.2.7.12</a> |
| 38            | 10NM         | G/O DISCRETE INPUT    | <a href="#">3.2.7.13</a> |
| 39            |              | G/O DISCRETE INPUT    | <a href="#">3.2.7.53</a> |
| 40            | FMS ALT      | G/O DISCRETE INPUT    | <a href="#">3.2.7.14</a> |
| 41            | ALT VOR      | G/O DISCRETE INPUT    | <a href="#">3.2.7.15</a> |
| 42            |              | G/O DISCRETE INPUT    | <a href="#">3.2.7.54</a> |
| 43            | VOR OFFSIDE  | G/O DISCRETE INPUT    | <a href="#">3.2.7.17</a> |

| EFI J2 Pinout |                |                      |                          |
|---------------|----------------|----------------------|--------------------------|
| J2            | SIGNAL NAME    | SIGNAL TYPE          | PARAGRAPH                |
| 44            | FMS OFFSIDE    | G/O DISCRETE INPUT   | <a href="#">3.2.7.16</a> |
| 45            | DIAG           | G/O DISCRETE INPUT   | <a href="#">3.2.7.69</a> |
| 46            | TFC            | G/O DISCRETE INPUT   | <a href="#">3.2.7.18</a> |
| 47            | CWX            | G/O DISCRETE INPUT   | <a href="#">3.2.7.19</a> |
| 48            | ARPT           | G/O DISCRETE INPUT   | <a href="#">3.2.7.20</a> |
| 49            | RTE DATA       | G/O DISCRETE INPUT   | <a href="#">3.2.7.21</a> |
| 50            | NAV AID        | G/O DISCRETE INPUT   | <a href="#">3.2.7.22</a> |
| 51            | 160NM          | G/O DISCRETE INPUT   | <a href="#">3.2.7.23</a> |
| 52            | 320NM          | G/O DISCRETE INPUT   | <a href="#">3.2.7.24</a> |
| 53            | 640NM          | G/O DISCRETE INPUT   | <a href="#">3.2.7.55</a> |
| 54            | CIWS RESET     | G/O DISCRETE INPUT   | <a href="#">3.2.7.25</a> |
| 55            | RAD ALT TEST   | G/O DISCRETE INPUT   | <a href="#">3.2.7.56</a> |
| 56            |                | G/O DISCRETE INPUT   | <a href="#">3.2.7.57</a> |
| 57            |                | 28V/O DISCRETE INPUT | <a href="#">3.2.7.58</a> |
| 58            |                | 28V/O DISCRETE INPUT | <a href="#">3.2.7.59</a> |
| 59            |                | REFERENCE blurg      | <a href="#">3.2.6.4</a>  |
| 60            |                | REFERENCE            | <a href="#">3.2.6.4</a>  |
| 61            | VOR REF H      | REFERENCE            | <a href="#">3.2.6.3</a>  |
| 62            | VOR REF L      | REFERENCE            | <a href="#">3.2.6.3</a>  |
| 63            | CHAS GND       | CHASSIS GROUND       | <a href="#">3.2.10.1</a> |
| 64            | 20NM           | G/O DISCRETE INPUT   | <a href="#">3.2.7.26</a> |
| 65            | 40NM           | G/O DISCRETE INPUT   | <a href="#">3.2.7.27</a> |
| 66            | 80NM           | G/O DISCRETE INPUT   | <a href="#">3.2.7.28</a> |
| 67            |                | 28V/O DISCRETE INPUT | <a href="#">3.2.7.29</a> |
| 68            |                | 28V/O DISCRETE INPUT | <a href="#">3.2.7.60</a> |
| 69            | LOC CAP        | 28V/O DISCRETE INPUT | <a href="#">3.2.7.30</a> |
| 70            | HDG SRC        | 28V/O DISCRETE INPUT | <a href="#">3.2.7.31</a> |
| 71            | FD ON          | G/O DISCRETE INPUT   | <a href="#">3.2.7.32</a> |
| 72            | SIG GND        | SIGNAL GROUND        | <a href="#">3.2.10.2</a> |
| 73            |                | 28V/O DISCRETE INPUT | <a href="#">3.2.7.61</a> |
| 74            | VERT DEV VALID | 28V/O DISCRETE INPUT | <a href="#">3.2.7.62</a> |
| 75            | EFIS VAL       | 28V/O DISCRETE INPUT | <a href="#">3.2.7.33</a> |
| 76            |                | 28V/O DISCRETE INPUT | <a href="#">3.2.7.34</a> |
| 77            |                | 28V/O DISCRETE INPUT | <a href="#">3.2.7.63</a> |
| 78            |                | 28V/O DISCRETE INPUT | <a href="#">3.2.7.64</a> |
| 79            | PMS/FMS        | 28V/O DISCRETE INPUT | <a href="#">3.2.7.65</a> |

## 3.1.2.3 EFI J3 Pinout

Table 3: EFI J3 Pinout

| EFI J3 Pinout |                  |                      |                          |
|---------------|------------------|----------------------|--------------------------|
| J3            | SIGNAL NAME      | SIGNAL TYPE          | PARAGRAPH                |
| 1             | PARITY           | G/O DISCRETE INPUT   | <a href="#">3.2.7.35</a> |
| 2             | AC 0             | G/O DISCRETE INPUT   | <a href="#">3.2.7.36</a> |
| 3             | LNAV BH (HS)     | 429 INPUT            | <a href="#">3.2.1.2</a>  |
| 4             | LNAV BL (HS)     | 429 INPUT            | <a href="#">3.2.1.2</a>  |
| 5             | LNAV AH (HS)     | 429 INPUT            | <a href="#">3.2.1.4</a>  |
| 6             | LNAV AL (HS)     | 429 INPUT            | <a href="#">3.2.1.4</a>  |
| 7             | LNAV CH (HS)     | 429 INPUT            | <a href="#">3.2.1.3</a>  |
| 8             | LNAV CL (HS)     | 429 INPUT            | <a href="#">3.2.1.3</a>  |
| 9             | INS A H (HS)     | 429 INPUT            | <a href="#">3.2.1.5</a>  |
| 10            | INS A L (HS)     | 429 INPUT            | <a href="#">3.2.1.5</a>  |
| 11            | EGPWS H          | ARINC 453 IN         | <a href="#">3.2.3.2</a>  |
| 12            | EGPWS L          | ARINC 453 IN         | <a href="#">3.2.3.2</a>  |
| 13            | EGPWS C          | ARINC 453 IN         | <a href="#">3.2.3.2</a>  |
| 14            | DME CLK H        | ARINC 568 IN         | <a href="#">3.2.4.1</a>  |
| 15            | DME CLK L        | ARINC 568 IN         | <a href="#">3.2.4.1</a>  |
| 16            | DME SYNC H       | ARINC 568 IN         | <a href="#">3.2.4.2</a>  |
| 17            | DME SYNC L       | ARINC 568 IN         | <a href="#">3.2.4.2</a>  |
| 18            | DME DAT H        | ARINC 568 IN         | <a href="#">3.2.4.3</a>  |
| 19            | DME DAT L        | ARINC 568 IN         | <a href="#">3.2.4.3</a>  |
| 20            | RANGE OUT H (LS) | ARINC 429 OUT        | <a href="#">3.2.12.1</a> |
| 21            | RANGE OUT L (LS) | ARINC 429 OUT        | <a href="#">3.2.12.1</a> |
| 22            | EGPWS OUT H (HS) | ARINC 429 OUT        | <a href="#">3.2.12.2</a> |
| 23            | EGPWS OUT L (HS) | ARINC 429 OUT        | <a href="#">3.2.12.2</a> |
| 24            | TTL              | 28V/O DISCRETE INPUT | <a href="#">3.2.7.66</a> |
| 25            |                  | G/O DISCRETE INPUT   | <a href="#">3.2.7.67</a> |
| 26            | CIWS TST         | G/O DISCRETE INPUT   | <a href="#">3.2.7.68</a> |
| 27            | PILOT/COPILOT    | G/O DISCRETE INPUT   | <a href="#">3.2.7.37</a> |
| 28            | ADI/HSI          | G/O DISCRETE INPUT   | <a href="#">3.2.7.38</a> |
| 29            | AC 4             | G/O DISCRETE INPUT   | <a href="#">3.2.7.42</a> |
| 30            | AC 5             | G/O DISCRETE INPUT   | <a href="#">3.2.7.43</a> |
| 31            | TCAS H (HS)      | 429 INPUT            | <a href="#">3.2.1.6</a>  |
| 32            | TCAS L (HS)      | 429 INPUT            | <a href="#">3.2.1.6</a>  |
| 33            | INS BH (HS)      | 429 INPUT            | <a href="#">3.2.1.7</a>  |
| 34            | INS BL (HS)      | 429 INPUT            | <a href="#">3.2.1.7</a>  |
| 35            | SPARE H (LS)     | 429 INPUT            | <a href="#">3.2.1.8</a>  |
| 36            | SPARE L (LS)     | 429 INPUT            | <a href="#">3.2.1.8</a>  |
| 37            | CWX H            | ARINC 453 IN         | <a href="#">3.2.3.1</a>  |
| 38            | CWX L            | ARINC 453 IN         | <a href="#">3.2.3.1</a>  |
| 39            | CWX C            | ARINC 453 IN         | <a href="#">3.2.3.1</a>  |
| 40            |                  | RS422 IN             | <a href="#">3.2.2.4</a>  |
| 41            |                  | RS422 IN             | <a href="#">3.2.2.4</a>  |
| 42            | XTLK 2H          | RS422 IN             | <a href="#">3.2.2.1</a>  |
| 43            | XTLK 2L          | RS422 IN             | <a href="#">3.2.2.1</a>  |

| EFI J3 Pinout |             |                    |                          |
|---------------|-------------|--------------------|--------------------------|
| J3            | SIGNAL NAME | SIGNAL TYPE        | PARAGRAPH                |
| 44            | XTLK 1H     | RS422 IN           | <a href="#">3.2.2.3</a>  |
| 45            | XTLK 1L     | RS422 IN           | <a href="#">3.2.2.3</a>  |
| 46            |             | RS422 IN           | <a href="#">3.2.2.2</a>  |
| 47            |             | RS422 IN           | <a href="#">3.2.2.2</a>  |
| 48            | AC 1        | G/O DISCRETE INPUT | <a href="#">3.2.7.39</a> |
| 49            | AC 2        | G/O DISCRETE INPUT | <a href="#">3.2.7.40</a> |
| 50            | AC 3        | G/O DISCRETE INPUT | <a href="#">3.2.7.41</a> |
| 51            |             |                    |                          |
| 52            |             |                    |                          |
| 53            |             |                    |                          |
| 54            | XTLK OUT H  | RS422 OUT          | <a href="#">3.2.13.1</a> |
| 55            | XTLK OUT L  | RS422 OUT          | <a href="#">3.2.13.1</a> |
| 56            | CHAS GND    | CHASSIS GROUND     | <a href="#">3.2.10.1</a> |
| 57            | XTLK OUT H  | RS422 OUT          | <a href="#">3.2.13.1</a> |
| 58            | XTLK OUT L  | RS422 OUT          | <a href="#">3.2.13.1</a> |
| 59            | DIAG OUT H  | RS422 OUT          | <a href="#">3.2.13.2</a> |
| 60            | DIAG OUT L  | RS422 OUT          | <a href="#">3.2.13.2</a> |
| 61            | DIAG IN H   | RS422 IN           | <a href="#">3.2.2.5</a>  |
| 62            | DIAG IN L   | RS422 IN           | <a href="#">3.2.2.5</a>  |
| 63            | SIG GND     | SIGNAL GROUND      | <a href="#">3.2.10.2</a> |
| 64            | TAWS H (LS) | ARINC 429 INPUT    | <a href="#">3.2.1.1</a>  |
| 65            | TAWS L (LS) | ARINC 429 INPUT    | <a href="#">3.2.1.1</a>  |
| 66            | SIG GND     | SIGNAL GROUND      | <a href="#">3.2.10.2</a> |
| 67            |             |                    |                          |
| 68            | XTLK OUT H  | RS422 OUT          | <a href="#">3.2.13.1</a> |
| 69            | XTLK OUT L  | RS422 OUT          | <a href="#">3.2.13.1</a> |
| 70            |             |                    |                          |
| 71            |             |                    |                          |
| 72            |             |                    |                          |
| 73            | ETI         | MAINT              | <a href="#">3.2.15.1</a> |
| 74            |             | G/O DISCRETE INPUT | <a href="#">3.2.7.75</a> |
| 75            |             | G/O DISCRETE INPUT | <a href="#">3.2.7.76</a> |
| 76            | SIG GND     | SIGNAL GROUND      | <a href="#">3.2.10.2</a> |
| 77            |             |                    |                          |
| 78            |             |                    |                          |
| 79            | ETI PWR     | MAINT              | <a href="#">3.2.15.2</a> |

### 3.1.2.4 Control Panel Pinout

Table 4 CP J1 Pinout

| Control Panel J1 Pinout |             |                     |                          |
|-------------------------|-------------|---------------------|--------------------------|
| J1                      | SIGNAL NAME | SIGNAL TYPE         | PARAGRAPH                |
| A                       | 5NM         | G/O DISCRETE OUTPUT | <a href="#">3.2.8.5</a>  |
| B                       | 10NM        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.6</a>  |
| C                       | 20NM        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.7</a>  |
| D                       | 40NM        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.8</a>  |
| E                       | 80NM        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.9</a>  |
| F                       | 160NM       | G/O DISCRETE OUTPUT | <a href="#">3.2.8.10</a> |
| G                       | 320NM       | G/O DISCRETE OUTPUT | <a href="#">3.2.8.11</a> |
| H                       | 640NM       | G/O DISCRETE OUTPUT | <a href="#">3.2.8.16</a> |
| J                       | 5VAC        | 5VAC POWER          | <a href="#">3.2.9.2</a>  |
| K                       | 5VAC RTN    | 5VAC RETURN         | <a href="#">3.2.9.2</a>  |
| L                       | CHSI        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.20</a> |
| M                       | MAP         | G/O DISCRETE OUTPUT | <a href="#">3.2.8.12</a> |
| N                       | FMS         | G/O DISCRETE OUTPUT | <a href="#">3.2.8.24</a> |
| P                       | PLAN        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.13</a> |
| R                       | SIG GND     | SIGNAL GROUND       | <a href="#">3.2.10.2</a> |
| S                       | DH UP       | G/O DISCRETE OUTPUT | <a href="#">3.2.8.14</a> |
| T                       | DH DOWN     | G/O DISCRETE OUTPUT | <a href="#">3.2.8.15</a> |
| U                       | TERR        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.23</a> |
| V                       | SIG GND     | SIGNAL GROUND       | <a href="#">3.2.10.2</a> |
| W                       | ARPT        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.17</a> |
| X                       | RTE DATA    | G/O DISCRETE OUTPUT | <a href="#">3.2.8.18</a> |
| Y                       | NAV AID     | G/O DISCRETE OUTPUT | <a href="#">3.2.8.19</a> |
| Z                       | CADI        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.25</a> |
| a                       | WYPT        | G/O DISCRETE OUTPUT | <a href="#">3.2.8.21</a> |
| b                       | CTR         | G/O DISCRETE OUTPUT | <a href="#">3.2.8.22</a> |
| c                       | 5VAC        | 5VAC POWER          | <a href="#">3.2.9.2</a>  |
| d                       |             |                     |                          |
| e                       | 5VAC RTN    | 5VAC RETURN         | <a href="#">3.2.9.2</a>  |
| f                       |             |                     |                          |
| g                       | SIG GND     | SIGNAL GROUND       | <a href="#">3.2.10.2</a> |
| h                       |             |                     |                          |
| i                       |             |                     |                          |
| j                       | SIG GND     | SIGNAL GROUND       | <a href="#">3.2.10.2</a> |
| k                       |             |                     |                          |
| m                       | TFC         | G/O DISCRETE OUTPUT | <a href="#">3.2.8.28</a> |
| n                       |             |                     |                          |
| p                       |             |                     |                          |
| q                       |             |                     |                          |
| r                       |             |                     |                          |
| s                       |             |                     |                          |
| t                       | VOR/ILS     | G/O DISCRETE OUTPUT | <a href="#">3.2.8.29</a> |

## 3.2 Electrical Interface Definition

Electrical interface definitions in the following paragraphs in many cases indicate sources and/or destinations of items outside the EFI Subsystem. These are based on the best information available and Astronautics understanding of the aircraft system and are not intended to supersede any Transaero or CMC system information for these items.

### 3.2.1 ARINC 429 Inputs

Notes for ARINC 429 Inputs:

Load: 12k ohm

Speeds:

High (HS) = 100kHz,

Low (LS) = 12.5kHz

Differential Voltages:

High = 6.5V to 13.0V,

Null = -2.5V to 2.5V,

Low = -13.0V to -6.5V

#### 3.2.1.1 TERRAIN AVOIDANCE and WARNING SYSTEM

Signal ID: TAWS H/L

Input/Output: Input

Signal Type: ARINC 429 LS

Wire: TSP

Function: Indicates the relative location of terrain with respect to the aircraft

Source: GPWS

| ADI         | HSI         |
|-------------|-------------|
| J3 - 64 = H | J3 - 64 = H |
| J3 - 65 = L | J3 - 65 = L |

#### 3.2.1.2 LONG RANGE NAVIGATION SOURCE B

Signal ID: LNAV B H/L

Input/Output: Input

Signal Type: ARINC 429 HS

Wire: TSP

Function: Provides navigation data

Source: FMS Offside

| ADI        | HSI        |
|------------|------------|
| J3 - 3 = H | J3 - 3 = H |
| J3 - 4 = L | J3 - 4 = L |

#### 3.2.1.3 LONG RANGE NAVIGATION SOURCE C

Signal ID: LNAV C H/L

Input/Output: Input

Signal Type: ARINC 429 HS

Wire: TSP

Function: Provides navigation data

Source: FMS Alternate

| ADI        | HSI        |
|------------|------------|
| J3 - 7 = H | J3 - 7 = H |
| J3 - 8 = L | J3 - 8 = L |

**3.2.1.4 LONG RANGE NAVIGATION SOURCE A**

Signal ID: LNAV A H/L  
 Input/Output: Input  
 Signal Type: ARINC 429 HS  
 Wire: TSP  
 Function: Provides navigation data  
 Source: FMS Onside

| ADI        | HSI        |
|------------|------------|
| J3 - 5 = H | J3 - 5 = H |
| J3 - 6 = L | J3 - 6 = L |

**3.2.1.5 INERTIAL NAVIGATION SYSTEM A**

Signal ID: INS A H/L  
 Input/Output: Input  
 Signal Type: ARINC 429 HS  
 Wire: TSP  
 Function: Provides navigation data  
 Source: INS Onside

| ADI         | HSI         |
|-------------|-------------|
| J3 - 9 = H  | J3 - 9 = H  |
| J3 - 10 = L | J3 - 10 = L |

**3.2.1.6 TRAFFIC COLLISION AVOIDANCE SYSTEM**

Signal ID: TCAS H/L  
 Input/Output: Input  
 Signal Type: ARINC 429 HS  
 Wire: TSP  
 Function: Provides traffic data  
 Source: TCAS

| HSI         |
|-------------|
| J3 - 31 = H |
| J3 - 32 = L |

**3.2.1.7 INERTIAL NAVIGATION SYSTEM B**

Signal ID: INS B H/L  
 Input/Output: Input  
 Signal Type: ARINC 429 HS  
 Wire: TSP  
 Function: Provides navigation data  
 Source: INS Alternate

| ADI         | HSI         |
|-------------|-------------|
| J3 - 33 = H | J3 - 33 = H |
| J3 - 34 = L | J3 - 34 = L |

**3.2.1.8 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: ARINC 429 LS  
 Wire: TSP  
 Function:  
 Source: \*\* Not used in this installation\*\*

|                            |
|----------------------------|
| J3 - 35 = H<br>J3 - 36 = L |
|----------------------------|

**3.2.2 RS422 Inputs**

Notes for RS422 Inputs:  
 Load: 4k ohm  
 Speed: 10MHz  
 Differential Voltages:  
 High = 2.0V to 6.0V  
 Low = -2.0V to -6.0V

**3.2.2.1 CROSSTALK 2**

Signal ID: XTLK 2 H/L  
 Input/Output: Input  
 Signal Type: RS422  
 Wire: TSP  
 Function: Comparison data from an EFI  
 Source: Offside HSI

| ADI                        | HSI                        |
|----------------------------|----------------------------|
| J3 - 42 = H<br>J3 - 43 = L | J3 - 42 = H<br>J3 - 43 = L |

**3.2.2.2 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: RS422  
 Wire: TSP  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|                            |
|----------------------------|
| J3 - 46 = H<br>J3 - 47 = L |
|----------------------------|

**3.2.2.3 CROSSTALK 1**

Signal ID: XTLK 1 H/L  
 Input/Output: Input  
 Signal Type: RS422  
 Wire: TSP  
 Function: Comparison data from an EFI  
 Source: Offside ADI

| ADI                        | HSI                        |
|----------------------------|----------------------------|
| J3 - 44 = H<br>J3 - 45 = L | J3 - 44 = H<br>J3 - 45 = L |

**3.2.2.4 (spare)**

Signal ID:  
 Input/Output: Input or Output  
 Signal Type: RS422  
 Wire: TSP  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|                            |
|----------------------------|
| J3 - 40 = H<br>J3 - 41 = L |
|----------------------------|

**3.2.2.5 DIAGNOSTIC IN**

Signal ID: DIAG IN H/L  
 Input/Output: Input  
 Signal Type: RS422  
 Wire: \*\*(Bench Use Only)\*\*  
 Function: Maintenance Purposes and Loading Software  
 Source: Channel Also Routed to Bezel Port

| ADI                        | HSI                        |
|----------------------------|----------------------------|
| J3 - 61 = H<br>J3 - 62 = L | J3 - 61 = H<br>J3 - 62 = L |

**3.2.3 ARINC 453 Inputs**

Note for ARINC 453 Inputs:  
 Load: 2k ohm (H,L)  
 78 ohm (H,C)  
 Speed: 1MHz  
 Voltage Range: 6.0V to -6.0V

**3.2.3.1 COLOR WEATHER RADAR**

Signal ID: CWX H/L/C  
 Input/Output: Input  
 Signal Type: ARINC 453  
 Wire: TST  
 Function: Weather Data  
 Source: \*\*(Not used in this installation)\*\*

|                                           |
|-------------------------------------------|
| J3 - 37 = H<br>J3 - 38 = L<br>J3 - 39 = C |
|-------------------------------------------|

**3.2.3.2 GROUND PROXIMITY WARNING SYSTEM**

Signal ID: EGPWS H/L/C  
 Input/Output: Input  
 Signal Type: ARINC 453  
 Wire: TSP  
 Function: Indicates the relative location of terrain with respect to the aircraft  
 Source: GPWS

|     |
|-----|
| HSI |
|-----|

|                                           |
|-------------------------------------------|
| J3 - 11 = H<br>J3 - 12 = L<br>J3 - 13 = C |
|-------------------------------------------|

### 3.2.4 ARINC 568 Inputs

Notes for ARINC 568 Inputs:

Load: 12k ohm

Speeds:

High = 100kHz,

Low = 12.5kHz

Differential Voltages:

High = 6.5V to 13.0V,

Null = -2.5V to 2.5V,

Low = -13.0V to -6.5V

#### 3.2.4.1 DME CLOCK

Signal ID: DME\_CLK H/L  
Input/Output: Input  
Signal Type: ARINC 568  
Wire: TSP  
Function: Distance Clock  
Source: DME

| ADI         | HSI         |
|-------------|-------------|
| J3 - 14 = H | J3 - 14 = H |
| J3 - 15 = L | J3 - 15 = L |

#### 3.2.4.2 DME SYNCHRONOUS

Signal ID: DME\_SYNC H/L  
Input/Output: Input  
Signal Type: ARINC 568  
Wire: TSP  
Function: Distance Sync  
Source: DME

| ADI         | HSI         |
|-------------|-------------|
| J3 - 16 = H | J3 - 16 = H |
| J3 - 17 = L | J3 - 17 = L |

#### 3.2.4.3 DME DATA

Signal ID: DME\_DAT H/L  
Input/Output: Input  
Signal Type: ARINC 568  
Wire: TSP  
Function: Distance Data  
Source: DME

| ADI         | HSI         |
|-------------|-------------|
| J3 - 18 = H | J3 - 18 = H |
| J3 - 19 = L | J3 - 19 = L |

### 3.2.5 Synchro Inputs

Notes for Synchro Inputs:  
 Range: 0 to 360 degrees,  
 Voltage: 11.8VAC  $\pm$  5%,  
 Frequency: 400Hz  $\pm$  20Hz

#### 3.2.5.1 SELECTED HEADING

Signal ID: SEL HDG X/Y/Z  
 Input/Output: Input  
 Signal Type: 11.8VAC/400Hz  
 Wire: TST  
 Function: Indicates the selected heading value with a pointer and a readout.  
 (0° = current heading, + = clockwise)  
 Source: Instrument Panel

| ADI         | HSI         |
|-------------|-------------|
| J1 - r = X  | J1 - r = X  |
| J1 - CC = Y | J1 - CC = Y |
| J1 - HH = Z | J1 - HH = Z |

#### 3.2.5.2 SELECTED COURSE

Signal ID: SEL CRS X/Y/Z  
 Input/Output: Input  
 Signal Type: 11.8VAC/400Hz  
 Wire: TST  
 Function: Indicates the selected course value with a pointer and a readout.  
 (0° = current heading, + = clockwise)  
 Source: Instrument Panel

| ADI        | HSI        |
|------------|------------|
| J1 - a = X | J1 - a = X |
| J1 - t = Y | J1 - t = Y |
| J1 - u = Z | J1 - u = Z |

#### 3.2.5.3 MAGNETIC HEADING

Signal ID: MAG HDG X/Y/Z  
 Input/Output: Input  
 Signal Type: 11.8VAC/400Hz  
 Wire: TST  
 Function: Magnetic Heading (0° = North, + = East of North)  
 Source: Compass Coupler

| ADI         | HSI         |
|-------------|-------------|
| J1 - s = X  | J1 - s = X  |
| J1 - DD = Y | J1 - DD = Y |
| J1 - Z = Z  | J1 - Z = Z  |

**3.2.5.4 VHF OMNIDIRECTIONAL RANGE RADIO**

Signal ID: VOR X/Y/Z  
 Input/Output: Input  
 Signal Type: 11.8VAC/400Hz  
 Wire: TST  
 Function: VOR Bearing Information (0° = nose of the aircraft, + = clockwise)  
 Source: VOR

| ADI         | HSI         |
|-------------|-------------|
| J1 - 13 = X | J1 - 13 = X |
| J1 - 12 = Y | J1 - 12 = Y |
| J1 - 11 = Z | J1 - 11 = Z |

**3.2.5.5 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: 11.8VAC/400Hz  
 Wire: TST  
 Function:  
 Source: \*\* Not used in this installation\*\*

|            |
|------------|
| J1 - S = X |
| J1 - T = Y |
| J1 - U = Z |

**3.2.5.6 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: 11.8VAC/400Hz  
 Wire: TST  
 Function:  
 Source: \*\* Not used in this installation\*\*

|             |
|-------------|
| J2 - 23 = X |
| J2 - 24 = Y |
| J2 - 25 = Z |

**3.2.5.7 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: 11.8VAC/400Hz  
 Wire: TST  
 Function:  
 Source: \*\* Not used in this installation\*\*

|             |
|-------------|
| J2 - 26 = X |
| J2 - 27 = Y |
| J2 - 28 = Z |

**3.2.6 Reference Inputs**

Note: for Reference Inputs:  
 Voltage 26VAC  $\pm$  10%,  
 Frequency 400Hz  $\pm$  20Hz

**3.2.6.1 SELECTED REFERENCE**

Signal ID: SEL REF H/L  
 Input/Output: Input  
 Signal Type: 26VAC/400Hz  
 Wire: TSP  
 Function: Selected Course/Heading and Magnetic Heading Synchro Reference  
 Source: A/C Reference Bus

| ADI        | HSI        |
|------------|------------|
| J1 - M = H | J1 - M = H |
| J1 - h = L | J1 - h = L |

**3.2.6.2 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: 26VAC/400Hz  
 Wire: TSP  
 Function:  
 Source: \*\* Not used in this installation\*\*

|                          |
|--------------------------|
| J1 - N = H<br>J1 - P = L |
|--------------------------|

**3.2.6.3 VOR REFERENCE**

Signal ID: VOR REF H/L  
 Input/Output: Input  
 Signal Type: 26VAC/400Hz  
 Wire: TSP  
 Function: VOR Synchro Reference  
 Source: AC Reference Bus

| ADI         | HSI         |
|-------------|-------------|
| J2 - 61 = H | J2 - 61 = H |
| J2 - 62 = L | J2 - 62 = L |

**3.2.6.4 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: 26VAC/400Hz  
 Wire: TSP  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|                            |
|----------------------------|
| J2 - 59 = H<br>J2 - 60 = L |
|----------------------------|

**3.2.7 Discrete Inputs**

Note for Discrete Inputs:  
 Ground = 0V to 3.5V,  
 Open = > 100k,  
 28V = 18.5V to 30V

**3.2.7.1 WEIGHT ON WHEELS**

Signal ID: WOW  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Gnd= Weight On Wheels, Open = not WOW  
 Source: Configuration Strapping

| ADI    | HSI    |
|--------|--------|
| J2 - 8 | J2 - 8 |

**3.2.7.2 CENTER**

Signal ID: CTR  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Gnd = Toggles between Center mode and Arc mode  
 Source: Control Panel

| ADI     | HSI     |
|---------|---------|
| J2 - 10 | J2 - 10 |

**3.2.7.3 HEADING VALID**

Signal ID: HDG VAL  
 Input/Output: Input  
 Signal Type: 28V/O  
 Wire: Single  
 Function: 28V = Magnetic Heading Valid, Open = Invalid  
 Source: Compass Coupler

| ADI     | HSI     |
|---------|---------|
| J2 - 14 | J2 - 14 |

**3.2.7.4 LOCALIZER VALID**

Signal ID: LOC VAL  
 Input/Output: Input  
 Signal Type: 28V/O  
 Wire: Single  
 Function: 28V = Localizer Valid, Open = Invalid  
 Source: VOR

| ADI     | HSI     |
|---------|---------|
| J2 - 17 | J2 - 17 |

**3.2.7.5 FLIGHT DIRECTOR VALID**

Signal ID: FD VAL  
 Input/Output: Input  
 Signal Type: 28V/O  
 Wire: Single  
 Function: 28V = Flight Director Valid, Open = Invalid  
 Source: Flight Director

| ADI | HSI |
|-----|-----|
|-----|-----|

J2 - 18

J2 - 18

**3.2.7.6 WAYPOINT**

Signal ID: WYPT  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Waypoints On and Off  
Source: Control Panel

ADI

HSI

J2 - 29

J2 - 29

**3.2.7.7 MAP MODE**

Signal ID: MAP  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Map Mode Selected  
Source: Control Panel

ADI

HSI

J2 - 31

J2 - 31

**3.2.7.8 FLIGHT MANAGEMENT SYSTEM MODE**

Signal ID: FMS  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: FMS Mode Selected  
Source: Control Panel

ADI

HSI

J2 - 32

J2 - 32

**3.2.7.9 PLAN MODE**

Signal ID: PLAN  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Plan Mode Selected  
Source: Control Panel

ADI

HSI

J2 - 33

J2 - 33

**3.2.7.10 DECISION HEIGHT UP**

Signal ID: DH UP  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Increase Decision Height Value  
Source: Control Panel

|         |
|---------|
| ADI     |
| J2 - 35 |

**3.2.7.11 DECISION HEIGHT DOWN**

Signal ID: DH DOWN  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Decrease Decision Height Value  
Source: Control Panel

|         |
|---------|
| ADI     |
| J2 - 36 |

**3.2.7.12 5 NAUTICAL MILE RANGE**

Signal ID: 5NM  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 5 Mile Range Selected  
Source: Control Panel

|         |
|---------|
| HSI     |
| J2 - 37 |

**3.2.7.13 10 NAUTICAL MILE RANGE**

Signal ID: 10NM  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 10 Mile Range Selected  
Source: Control Panel

|         |
|---------|
| HSI     |
| J2 - 38 |

**3.2.7.14 FMS ALTERNATE**

Signal ID: FMS ALT  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Used in conjunction with FMS OFFSIDE 3.2.7.16,  
 FMS ALT = FMS OFFSIDE = Open (FMS Source = Onside)  
 FMS ALT = d/c, FMS OFFSIDE = Gnd (FMS Source = Offside)  
 FMS ALT = Gnd, FMS OFFSIDE = Open (FMS Source = Alternate)  
 Source: Instrument Transfer Panel

| ADI     | HSI     |
|---------|---------|
| J2 - 40 | J2 - 40 |

**3.2.7.15 ALTERNATE VOR**

Signal ID: ALT VOR  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Used in conjunction with VOR OFFSIDE 3.2.7.17,  
 ALT VOR = Open, VOR OFFSIDE = Open (Radio Source = Onside Radio)  
 ALT VOR = Gnd, VOR OFFSIDE = d/c (Radio Source = Alternate Radio)  
 ALT VOR = Open, VOR OFFSIDE = Gnd (Radio Source = Offside Radio)  
 Source: Instrument Transfer Panel

| ADI     | HSI     |
|---------|---------|
| J2 - 41 | J2 - 41 |

**3.2.7.16 FMS OFFSIDE**

Signal ID: FMS OFFSIDE  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Used in conjunction with FMS ALTERNATE 3.2.7.14,  
 FMS ALT = FMS OFFSIDE = Open (FMS Source = Onside)  
 FMS ALT = d/c, FMS OFFSIDE = Gnd (FMS Source = Offside)  
 FMS ALT = Gnd, FMS OFFSIDE = Open (FMS Source = Alternate)  
 Source: Instrument Transfer Panel

| ADI     | HSI     |
|---------|---------|
| J2 - 44 | J2 - 44 |

**3.2.7.17 VOR OFFSIDE**

Signal ID: VOR OFFSIDE  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Used in conjunction with VOR ALTERNATE 3.2.7.15,  
 ALT VOR = Open, VOR OFFSIDE = Open (Radio Source = Onside Radio)  
 ALT VOR = Gnd, VOR OFFSIDE = d/c (Radio Source = Alternate Radio)  
 ALT VOR = Open, VOR OFFSIDE = Gnd (Radio Source = Offside Radio)  
 Source: Instrument Transfer Panel

| ADI | HSI |
|-----|-----|
|-----|-----|

J2 - 43

J2 - 43

**3.2.7.18TRAFFIC**

Signal ID: TFC  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Traffic Overlay On and Off  
Source: Control Panel

HSI

J2 - 46

**3.2.7.19WEATHER**

Signal ID: CWX  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Weather On and Off  
Source: \*\*(Not used in this installation)\*\*

J2 - 47

**3.2.7.20AIRPORT**

Signal ID: ARPT  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Airports On and Off  
Source: Control Panel

ADI

J2 - 48

HSI

J2 - 48

**3.2.7.21 ROUTE DATA**

Signal ID: RTE DATA  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Route Data On and Off  
Source: Control Panel

ADI

J2 - 49

HSI

J2 - 49

**3.2.7.22 NAVIGATION AID STATION**

Signal ID: NAV AID  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Stations On and Off  
Source: Control Panel

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 50 | J2 - 50 |

**3.2.7.23 160 NAUTICAL MILE RANGE**

Signal ID: 160NM  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 160 Mile Range Selected  
Source: Control Panel

|         |
|---------|
| HSI     |
| J2 - 51 |

**3.2.7.24 320 NAUTICAL MILE RANGE**

Signal ID: 320NM  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 320 Mile Range Selected  
Source: Control Panel

|         |
|---------|
| HSI     |
| J2 - 52 |

**3.2.7.25 CENTRAL INSTRUMENT WARNING SYSTEM RESET**

Signal ID: CIWS RESET  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Clear CIWS alerts  
Source: Central Instrument Warning System

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 54 | J2 - 54 |

**3.2.7.26 20 NAUTICAL MILE RANGE**

Signal ID: 20NM  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 20 Mile Range Selected  
Source: Control Panel

|     |
|-----|
| HSI |
|-----|

|         |
|---------|
| J2 - 64 |
|---------|

**3.2.7.2740 NAUTICAL MILE RANGE**

Signal ID: 40NM  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 40 Mile Range Selected  
Source: Control Panel

|     |
|-----|
| HSI |
|-----|

|         |
|---------|
| J2 - 65 |
|---------|

**3.2.7.28 80 NAUTICAL MILE RANGE**

Signal ID: 80NM  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 80 Mile Range Selected  
Source: Control Panel

|     |
|-----|
| HSI |
|-----|

|         |
|---------|
| J2 - 66 |
|---------|

**3.2.7.29(spare)**

Signal ID:  
Input/Output: Input  
Signal Type: 28V/Open  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 67 |
|---------|

**3.2.7.30 LOCALIZER CAPTURED**

Signal ID: LOC CAP  
Input/Output: Input  
Signal Type: 28V/Open  
Wire: Single  
Function: Gnd = Localizer Frequency Captured  
Source: Flight Director

|     |
|-----|
| ADI |
|-----|

|         |
|---------|
| J2 - 69 |
|---------|

|     |
|-----|
| HSI |
|-----|

|         |
|---------|
| J2 - 69 |
|---------|

**3.2.7.31 HEADING SOURCE**

Signal ID: HDG SRC  
 Input/Output: Input  
 Signal Type: 28V/Open  
 Wire: Single  
 Function: 28V = Alternate Heading Source Selected  
 Source: Instrument Transfer Panel

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 70 | J2 - 70 |

**3.2.7.32 FLIGHT DIRECTOR ON**

Signal ID: FD ON  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Gnd = Flight Director On  
 Source: \*\*(Strap to Ground)\*\*

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 71 | J2 - 71 |

**3.2.7.33 ELECTRONIC FLIGHT INSTRUMENT SYSTEM VALID**

Signal ID: EFIS VAL  
 Input/Output: Input  
 Signal Type: 28V/Open  
 Wire: Single  
 Function: 28V = EFI Valid, Open = Invalid  
 Source: Onside Electronic Flight Instrument

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 75 | J2 - 75 |

**3.2.7.34 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: 28V/Open  
 Wire: Single  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 76 |
|---------|

**3.2.7.35 PARITY**

Signal ID: Parity  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Use to create odd configuration total  
 Source: Configuration Strapping, see Table 5

|        |        |
|--------|--------|
| ADI    | HSI    |
| J3 - 1 | J3 - 1 |

**3.2.7.36 AIRCRAFT CONFIGURATION 0**

Signal ID: AC 0  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: AC0-AC6 combination configures EFI to A/C  
 Source: Configuration Strapping, see Table 5

| ADI    | HSI    |
|--------|--------|
| J3 - 2 | J3 - 2 |

**3.2.7.37 PILOT/COPILOT**

Signal ID: PILOT/COPILOT  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Gnd = Copilot, Open = Pilot  
 Source: Configuration Strapping, see Table 5

| ADI     | HSI     |
|---------|---------|
| J3 - 27 | J3 - 27 |

**3.2.7.38ADI / HSI**

Signal ID: ADI/HSI  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Gnd = HSI, Open = ADI  
 Source: Configuration Strapping, see Table 5

| ADI     | HSI     |
|---------|---------|
| J3 - 28 | J3 - 28 |

**3.2.7.39 AIRCRAFT CONFIGURATION 1**

Signal ID: AC 1  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: AC0-AC6 combination configures EFI to A/C  
 Source: Configuration Strapping, see Table 5

| ADI     | HSI     |
|---------|---------|
| J3 - 48 | J3 - 48 |

**3.2.7.40 AIRCRAFT CONFIGURATION 2**

Signal ID: AC 2  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: AC0-AC6 combination configures EFI to A/C  
 Source: Configuration Strapping, see Table 5

| ADI     | HSI     |
|---------|---------|
| J3 - 49 | J3 - 49 |

**3.2.7.41 AIRCRAFT CONFIGURATION 3**

Signal ID: AC 3  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: AC0-AC6 combination configures EFI to A/C  
 Source: Configuration Strapping, see Table 5

| ADI     | HSI     |
|---------|---------|
| J3 - 50 | J3 - 50 |

**3.2.7.42 AIRCRAFT CONFIGURATION 4**

Signal ID: AC 4  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: AC0-AC6 combination configures EFI to A/C  
 Source: Configuration Strapping, see Table 5

| ADI     | HSI     |
|---------|---------|
| J3 - 29 | J3 - 29 |

**3.2.7.43 AIRCRAFT CONFIGURATION 5**

Signal ID: AC 5  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: AC0-AC6 combination configures EFI to A/C  
 Source: Configuration Strapping, see Table 5

| ADI     | HSI     |
|---------|---------|
| J3 - 30 | J3 - 30 |

**3.2.7.44 AIRCRAFT CONFIGURATION 6**

Signal ID: AC 6  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: AC0-AC6 combination configures EFI to A/C  
 Source: Configuration Strapping, see Table 5

| ADI | HSI |
|-----|-----|
|-----|-----|

J2 - 7

J2 - 7

**3.2.7.45 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: 28V/O  
 Wire: Single  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

J1 - A

**3.2.7.46 INERTIAL REFERENCE UNIT ALTERNATE**

Signal ID: IRU ALT  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Open = IRU Onside, Gnd = IRU Alternate  
 Source: Instrument Transfer Panel

|        |        |
|--------|--------|
| ADI    | HSI    |
| J2 - 9 | J2 - 9 |

**3.2.7.47 GLIDESLOPE VALID**

Signal ID: G/S VAL  
 Input/Output: Input  
 Signal Type: 28V/O  
 Wire: Single  
 Function: 28V = Glideslope Valid, Open = Invalid  
 Source: VOR

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 15 | J2 - 15 |

**3.2.7.48 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: 28V/O  
 Wire: Single  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

J2 - 16

**3.2.7.49 SPEED DEVIATION VALID**

Signal ID: SPD DEV VAL  
 Input/Output: Input  
 Signal Type: 28V/O  
 Wire: Single  
 Function: 28V = Speed Deviation Valid, Open = Invalid  
 Source: PMS/Auto-throttle

|     |     |
|-----|-----|
| ADI | HSI |
|-----|-----|

J2 - 19

J2 - 19

**3.2.7.50 RADIO ALTIMETER VALID**

Signal ID: RAD ALT VAL  
 Input/Output: Input  
 Signal Type: 28V/O  
 Wire: Single  
 Function: 28V = RAD ALT Valid, Open = Invalid  
 Source: Radio Altimeter

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 20 | J2 - 20 |

**3.2.7.51 COMBINED MODE**

Signal ID: COMB  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Gnd = Combined Mode  
 Source: Control Panel

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 30 | J2 - 30 |

**3.2.7.52 TERRAIN**

Signal ID: TERR  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function: Gnd = Toggles Terrain Mode On and Off  
 Source: Control Panel

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 34 | J2 - 34 |

**3.2.7.53 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

J2 - 39

**3.2.7.54 (spare)**

Signal ID:  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 42 |
|---------|

**3.2.7.55 640 NM**

Signal ID: 640  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 640 Mile Range Selected  
Source: Control Panel

|         |
|---------|
| HSI     |
| J2 - 53 |

**3.2.7.56 RADIO ALTIMETER TEST**

Signal ID: RAD ALT TEST  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Test Mode, Open = No Test  
Source: Radio Altimeter

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 55 | J2 - 55 |

**3.2.7.57(spare)**

Signal ID:  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 56 |
|---------|

**3.2.7.58 (spare)**

Signal ID:  
Input/Output: Input  
Signal Type: 28V/Open  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 57 |
|---------|

**3.2.7.59 (spare)**

Signal ID:  
Input/Output: Input  
Signal Type: 28V/Open  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 58 |
|---------|

**3.2.7.60 (spare)**

Signal ID:  
Input/Output: Input  
Signal Type: 28V/Open  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 68 |
|---------|

**3.2.7.61 (spare)**

Signal ID:  
Input/Output: Input  
Signal Type: 28V/Open  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 73 |
|---------|

**3.2.7.62 VERTICAL DEVIATION VALID**

Signal ID: VERT DEV VAL  
Input/Output: Input  
Signal Type: 28V/O  
Wire: Single  
Function: 28V = Vertical Deviation Valid, Open = Invalid  
Source: PMS

|     |
|-----|
| HSI |
|-----|

|         |
|---------|
| J2 - 74 |
|---------|

**3.2.7.63 (spare)**

Signal ID:  
Input/Output: Input  
Signal Type: 28V/Open  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 77 |
|---------|

**3.2.7.64(spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: 28V/Open  
 Wire: Single  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J2 - 78 |
|---------|

**3.2.7.65PMS/FMS**

Signal ID: PMS/FMS  
 Input/Output: Input  
 Signal Type: 28V/Open  
 Wire: Single  
 Function: 28V=PMS, Open=FMS  
 Source:

|         |
|---------|
| HSI     |
| J2 - 79 |

**3.2.7.66 TUNED TO LOCALIZER**

Signal ID: TTL  
 Input/Output: Input  
 Signal Type: 28V/O  
 Wire: Single  
 Function: 28V = TTL, Open = not TTL  
 Source: VOR

| ADI     | HSI     |
|---------|---------|
| J3 - 24 | J3 - 24 |

**3.2.7.67 (spare)**

Signal ID:  
 Input/Output: Input  
 Signal Type: G/Open  
 Wire: Single  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J3 - 25 |
|---------|

**3.2.7.68 CENTRAL INSTRUMENT WARNING SYSTEM TEST**

Signal ID: CIWS TST  
 Input/Output: Input  
 Signal Type: G/Open  
 Wire: Single  
 Function: Gnd = Test Mode, Open = No Test  
 Source: CIWS

| ADI     | HSI     |
|---------|---------|
| J3 - 26 | J3 - 26 |

**3.2.7.69 DIAGNOSTIC**

Signal ID: DIAG  
Input/Output: Input  
Signal Type: G/O  
Wire: \*\*(Bench Use Only)\*\*  
Function: Gnd = Diagnostic Mode, Open = Flight Mode  
Maintenance Purposes and Loading Software  
Source: Configuration Strapping  
Note: Diagnostic Mode, Also enables Bezel Port

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 45 | J2 - 45 |

**3.2.7.70 (provisional)**

Signal ID:  
Input/Output: Input  
Signal Type: 28V/Open  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|        |
|--------|
| J1 – B |
|--------|

**3.2.7.71 (spare)**

Signal ID:  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|        |
|--------|
| J1 – W |
|--------|

**3.2.7.72(spare)**

Signal ID:  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|        |
|--------|
| J1 – p |
|--------|

**3.2.7.73 (spare)**

Signal ID:  
Input/Output: Input  
Signal Type: G/O  
Wire: Single  
Function:  
Source: \*\*(Not used in this installation)\*\*

|        |
|--------|
| J1 – q |
|--------|

**3.2.7.74 (provisional)**

Signal ID:  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J1 – BB |
|---------|

**3.2.7.75 (provisional)**

Signal ID:  
 Input/Output: Input  
 Signal Type: G/Open  
 Wire: Single  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J3 - 74 |
|---------|

**3.2.7.76 (provisional)**

Signal ID:  
 Input/Output: Input  
 Signal Type: G/O  
 Wire: Single  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|         |
|---------|
| J3 - 75 |
|---------|

**3.2.8 Discrete Outputs**

Note: for Discrete Outputs:  
 Ground/Open: sink 150mA,  
 28V/Open: source 160mA,  
 Ground = 0V to 3.5V,  
 Open = > 100k,  
 28V = 18.5V to 30V

**3.2.8.1 EFIS VALID OUT**

Signal ID: EFIS VAL OUT  
 Input/Output: Output  
 Signal Type: 28V/Open  
 Wire: Single  
 Function: 28V = Electronic Flight Instrument Valid  
 Destination: Onside EFI

|        |        |
|--------|--------|
| ADI    | HSI    |
| J2 - 5 | J2 - 5 |

**3.2.8.2 MASTER LAMP**

Signal ID: MSTR LMP  
Input/Output: Output  
Signal Type: 28V/Open  
Wire: Single  
Function: 28V = Master Lamp On, Open = Lamp Off  
Destination: Central Instrument Warning System

|        |        |
|--------|--------|
| ADI    | HSI    |
| J2 - 6 | J2 - 6 |

**3.2.8.3 ATTITUDE LAMP**

Signal ID: ATT LMP  
Input/Output: Output  
Signal Type: 28V/Open  
Wire: Single  
Function: 28V = Attitude Miscompare On, Open = Miscompare Off  
Destination: Central Instrument Warning System

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 21 | J2 - 21 |

**3.2.8.4 HEADING LAMP**

Signal ID: HDG LMP  
Input/Output: Output  
Signal Type: 28V/Open  
Wire: Single  
Function: 28V = Heading Miscompare On, Open = Miscompare Off  
Destination: Central Instrument Warning System

|         |         |
|---------|---------|
| ADI     | HSI     |
| J2 - 22 | J2 - 22 |

**3.2.8.5 5 NAUTICAL MILE RANGE**

Signal ID: 5NM  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 5 Mile Range Selected  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - A |

**3.2.8.6 10 NAUTICAL MILE RANGE**

Signal ID: 10NM  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 10 Mile Range Selected  
Destination: EFI

|    |
|----|
| CP |
|----|

|        |
|--------|
| J1 - B |
|--------|

**3.2.8.7 20 NAUTICAL MILE RANGE**

Signal ID: 20NM  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 20 Mile Range Selected  
Destination: EFI

|    |
|----|
| CP |
|----|

|        |
|--------|
| J1 - C |
|--------|

**3.2.8.8 40 NAUTICAL MILE RANGE**

Signal ID: 40NM  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 40 Mile Range Selected  
Destination: EFI

|    |
|----|
| CP |
|----|

|        |
|--------|
| J1 - D |
|--------|

**3.2.8.9 80 NAUTICAL MILE RANGE**

Signal ID: 80NM  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 80 Mile Range Selected  
Destination: EFI

|    |
|----|
| CP |
|----|

|        |
|--------|
| J1 - E |
|--------|

**3.2.8.10 160 NAUTICAL MILE RANGE**

Signal ID: 160NM  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 160 Mile Range Selected  
Destination: EFI

|    |
|----|
| CP |
|----|

|        |
|--------|
| J1 - F |
|--------|

**3.2.8.11 320 NAUTICAL MILE RANGE**

Signal ID: 320NM  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 320 Mile Range Selected  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - G |

**3.2.8.12 MAP MODE**

Signal ID: MAP  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = MAP Mode Selected  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - M |

**3.2.8.13 PLAN MODE**

Signal ID: PLAN  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = PLAN Mode Selected  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - P |

**3.2.8.14DECISION HEIGHT UP**

Signal ID: DH UP  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = DH Increase  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - S |

**3.2.8.15DECISION HEIGHT DOWN**

Signal ID: DH DOWN  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = DH Decrease  
Destination: EFI

|    |
|----|
| CP |
|----|

|        |
|--------|
| J1 - T |
|--------|

**3.2.8.16 640 NAUTICAL MILE RANGE**

Signal ID: 640NM  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = 640 Mile Range Selected  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - H |

**3.2.8.17 AIRPORT**

Signal ID: ARPT  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Airports On and Off  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - W |

**3.2.8.18 ROUTE DATA**

Signal ID: RTE DATA  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Route Data On and Off  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - X |

**3.2.8.19 NAVIGATION AID**

Signal ID: NAV AID  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Stations On and Off  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - Y |

**3.2.8.20 COMBINED HSI**

Signal ID: CHSI  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = HSI Combined Mode Selected  
Destination: HSI

|        |
|--------|
| CP     |
| J1 – L |

**3.2.8.21 WAYPOINT**

Signal ID: WYPT  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Waypoints On and Off  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - a |

**3.2.8.22 CENTER**

Signal ID: CTR  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Arc and Center Display  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - b |

**3.2.8.23 TERRAIN**

Signal ID: TERR  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Terrain Mode On and Off  
Destination: EFI

|        |
|--------|
| CP     |
| J1 - U |

**3.2.8.24 FLIGHT MANAGEMENT SYSTEM**

Signal ID: FMS  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = FMS Selected  
Destination: EFI

|    |
|----|
| CP |
|----|

|        |
|--------|
| J1 - N |
|--------|

**3.2.8.25 COMBINED ADI**

Signal ID: CADI  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = ADI Combined Mode Selected  
Destination: ADI

|    |
|----|
| CP |
|----|

|        |
|--------|
| J1 - Z |
|--------|

**3.2.8.26 (spare)**

Signal ID:  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function:  
Destination: \*\*(Not used in this installation)\*\*

|        |
|--------|
| J1 - G |
|--------|

**3.2.8.27 (spare)**

Signal ID:  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function:  
Destination: \*\*(Not used in this installation)\*\*

|        |
|--------|
| J1 - H |
|--------|

**3.2.8.28 TRAFFIC**

Signal ID: TFC  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function: Gnd = Toggles Traffic On and Off  
Destination: EFI

|    |
|----|
| CP |
|----|

|        |
|--------|
| J1 - m |
|--------|

**3.2.8.29 VOR/ILS**

Signal ID:  
Input/Output: Output  
Signal Type: G/O  
Wire: Single  
Function:  
Destination: \*\*(No connection required to EFI)\*\*

|    |
|----|
| CP |
|----|

### 3.2.9 Power Inputs

#### 3.2.9.1 28VDC POWER

Signal ID: 28VDC/RTN  
 Input/Output: Input  
 Signal Type: 28VDC input power per DO-160D, Sect 16, Cat Z  
 Wire: TP  
 Function: EFI Input Power  
 Source: A/C Power Bus

| ADI                              | HSI                              |
|----------------------------------|----------------------------------|
| J1 - C,D,Y = H<br>J1 - E,F,b = L | J1 - C,D,Y = H<br>J1 - E,F,b = L |

#### 3.2.9.2 5VAC LIGHTING POWER

Signal ID: 5VAC/RTN  
 Input/Output: Input  
 Signal Type: 5VAC input lighting power  
 Wire: TP  
 Function: EFI Bezel Lighting Power  
 Source: A/C Lighting Power Bus

| ADI                      | HSI                      | CP                           |
|--------------------------|--------------------------|------------------------------|
| J1 - K = H<br>J1 - L = L | J1 - K = H<br>J1 - L = L | J1 - J,c = H<br>J1 - K,e = L |

### 3.2.10 Grounds

#### 3.2.10.1 CHASSIS GROUND

Signal ID: CHAS GND  
 Input/Output: Output  
 Signal Type: Chassis Connection  
 Wire: Single  
 Function: Electrical connection providing chassis tie  
 Destination: Mating connector back shell

| ADI                          | HSI                          |
|------------------------------|------------------------------|
| J1 - X<br>J2 - 63<br>J3 - 56 | J1 - X<br>J2 - 63<br>J3 - 56 |

#### 3.2.10.2 SIGNAL GROUND

Signal ID: SIG GND  
 Input/Output: Output  
 Signal Type: Signal Ground Reference  
 Wire: Single  
 Function: Electrical connection providing signal ground reference  
 Destination: A/C ground

| ADI                | HSI     | CP           |
|--------------------|---------|--------------|
| J1 - GG<br>J2 - 72 | J1 - GG | J1 - R,V,g,j |

|               |               |  |
|---------------|---------------|--|
| J3 – 63,66,76 | J2 - 72       |  |
|               | J3 – 63,66,76 |  |

**3.2.10.3 GROUND**

Signal ID: CIWS RTN  
Input/Output: Output  
Signal Type: Ground Reference  
Wire: Single  
Function: Electrical connection providing ground reference. Internally connected to 28vdc return through common mode choke.  
Destination: CIWS

| ADI    | HSI    |
|--------|--------|
| J1 - V | J1 - V |

**3.2.11 Analog Inputs****3.2.11.1 TO/FROM**

Signal ID: T/F H/L  
Input/Output: Input  
Signal Type: Analog  
Wire: TSP  
Function: Indicates whether the aircraft is flying toward or away from the selected radio navigational aid  
Range:  $\pm 0.040V$   
Load:  $1k\Omega$   
Source: VOR

| ADI        | HSI        |
|------------|------------|
| J1 - c = H | J1 - c = H |
| J1 - d = L | J1 - d = L |

**3.2.11.2 LOCALIZER (CRS)**

Signal ID: LOC (CRS) H/L  
Input/Output: Input  
Signal Type: Analog  
Wire: TSP  
Function: Indicates the relative error in the horizontal plane between the desired aircraft track and its actual position  
(left of center = right of the track)  
Range:  $\pm 0.150V$   
Load:  $2k\Omega$  (Localizer CRS and Localizer FINE must be connected to make  $1k\Omega$  input impedance)  
Source: VOR

| ADI        | HSI        |
|------------|------------|
| J1 - m = H | J1 - m = H |
| J1 - z = L | J1 - z = L |

**3.2.11.3 LOCALIZER (FINE)**

Signal ID: LOC (FINE) H/L  
 Input/Output: Input  
 Signal Type: Analog  
 Wire: TSP  
 Function: Indicates the relative error in the horizontal plane between the desired aircraft track and its actual position  
 (left of center = right of the track)  
 Range:  $\pm 0.045V$  (used for expanded scale mode)  
 Load: 2k $\Omega$  (Localizer CRS and Localizer FINE must be connected to make 1k $\Omega$  input impedance)  
 Source: VOR

| ADI         | HSI         |
|-------------|-------------|
| J1 - n = H  | J1 - n = H  |
| J1 - AA = L | J1 - AA = L |

**3.2.11.4 SPEED DEVIATION**

Signal ID: SPD DEV H/L  
 Input/Output: Input  
 Signal Type: Analog  
 Wire: TSP  
 Function: Relative error between the suggested speed and the aircraft's speed (+ = too fast)  
 Range:  $\pm 2.2V$   
 Load: 2k $\Omega$   
 Source: PMS/Auto-throttle

| ADI        | HSI        |
|------------|------------|
| J1 - i = H | J1 - i = H |
| J1 - j = L | J1 - j = L |

**3.2.11.5 ROLL COMMAND**

Signal ID: ROLL CMD H/L  
 Input/Output: Input  
 Signal Type: Analog  
 Wire: TSP  
 Function: Relative error between the suggested roll and the aircraft's roll (+ = commanding roll clockwise)  
 Range:  $\pm 0.500V$   
 Load: 2k $\Omega$   
 Source: Flight Director

| ADI        | HSI        |
|------------|------------|
| J1 - g = H | J1 - g = H |
| J1 - x = L | J1 - x = L |

**3.2.11.6 VERTICAL DEVIATION**

Signal ID: VERT DEV H/L  
 Input/Output: Input  
 Signal Type: Analog  
 Wire: TSP  
 Function: Indicates the relative error in the vertical plane between the flight plan altitude and the aircraft's position.  
 (above center = aircraft is below the required vertical position)  
 Range:  $\pm 0.150V$   
 Load:  $2k\Omega$   
 Source: Performance Management System

| ADI         | HSI         |
|-------------|-------------|
| J1 - y = H  | J1 - y = H  |
| J1 - FF = L | J1 - FF = L |

**3.2.11.7 PITCH COMMAND**

Signal ID: PITCH CMD H/L  
 Input/Output: Input  
 Signal Type: Analog  
 Wire: TSP  
 Function: Relative error between the suggested roll and the aircraft's roll  
 (+ = commanding pitch up)  
 Range:  $\pm 0.500V$   
 Load:  $2k\Omega$   
 Source: Flight Director

| ADI        | HSI        |
|------------|------------|
| J2 - 3 = H | J2 - 3 = H |
| J2 - 4 = L | J2 - 4 = L |

**3.2.11.8 RADIO ALTIMETER**

Signal ID: RAD ALT H/L  
 Input/Output: Input  
 Signal Type: Analog  
 Wire: TSP  
 Function: Displays the height above ground  
 Range:  $26.42V$   
 Load:  $20k\Omega$   
 Source: Radio Altimeter

| ADI        | HSI        |
|------------|------------|
| J2 - 1 = H | J2 - 1 = H |
| J2 - 2 = L | J2 - 2 = L |

**3.2.11.9 GLIDESLOPE**

Signal ID: GS H/L  
 Input/Output: Input  
 Signal Type: Analog  
 Wire: TSP  
 Function: Indicates the relative error in the vertical plane between the glideslope approach path and the aircraft's position.  
 (Above center = aircraft is below the required vertical position)  
 Range:  $\pm 0.150V$   
 Load:  $1k\Omega$   
 Source: VOR

| ADI        | HSI        |
|------------|------------|
| J1 - R = H | J1 - R = H |
| J1 - k = L | J1 - k = L |

**3.2.12 ARINC 429 Outputs**

Notes for ARINC 429 Outputs

Load: 12k ohm minimum

Speeds: High (HS) = 100kHz,

Low (LS) = 12.5kHz

Differential Voltages:

High = 6.5V to 13.0V,

Null = -2.5V to 2.5V,

Low = -13.0V to -6.5V

**3.2.12.1 RANGE OUT**

Signal ID: RANGE OUT H/L  
 Input/Output: Output  
 Signal Type: ARINC 429 LS  
 Wire: TSP  
 Function: Outputs range data  
 Destination: FMS and GPWS

| HSI         |
|-------------|
| J3 - 20 = H |
| J3 - 21 = L |

**3.2.12.2 EGPWS OUT**

Signal ID: EGPWS OUT H/L  
 Input/Output: Output  
 Signal Type: ARINC 429 HS  
 Wire: TSP  
 Function: Outputs Ground Proximity Warning System data  
 Destination: GPWS

| ADI         | HSI         |
|-------------|-------------|
| J3 - 22 = H | J3 - 22 = H |
| J3 - 23 = L | J3 - 23 = L |

### 3.2.13 RS422 Outputs

Notes for RS422 Outputs

Load: 4k ohm

Speed: 10MHz

Differential Voltages:

High = 2.0V to 6.0V,

Low = -2.0V to -6.0V

#### 3.2.13.1 CROSSTALK OUT

Signal ID: XTLK\_OUT H/L

Input/Output: Output

Signal Type: RS422

Wire: TSP

Function: Data to other EFIs

Destination: EFIs

| ADI               | HSI               |
|-------------------|-------------------|
| J3 – 54,57,68 = H | J3 – 54,57,68 = H |
| J3 – 55,58,69 = L | J3 – 55,58,69 = L |

#### 3.2.13.2 DIAGNOSTIC OUT

Signal ID: DIAG OUT H/L

Input/Output: Output

Signal Type: RS422

Wire: \*\*(Bench Use Only)\*\*

Function: Maintenance Purposes and Loading Software

Source: Channel Also Routed to Bezel Port

| ADI         | HSI         |
|-------------|-------------|
| J3 – 59 = H | J3 – 59 = H |
| J3 – 60 = L | J3 – 60 = L |

### 3.2.14 Analog Outputs

#### 3.2.14.1 (spare)

Signal ID:

Input/Output: Output

Signal Type: Analog

Wire: TSP

Function:

Destination: \*\*(Not used in this installation)\*\*

|            |
|------------|
| J1 - e = H |
| J1 - f = L |

**3.2.14.2 (spare)**

Signal ID:  
 Input/Output: Output  
 Signal Type: Analog  
 Wire: TSP  
 Function:  
 Source: \*\*(Not used in this installation)\*\*

|                          |
|--------------------------|
| J1 - v = H<br>J1 - w = L |
|--------------------------|

**3.2.15 Miscellaneous****3.2.15.1 ELAPSED TIME INDICATOR**

Signal ID: ETI  
 Input/Output: Output  
 Signal Type: Serial Data  
 Function: Data for cumulative time of applied power to EFI  
 Note: \*\*(Bench Use Only)\*\* with ETI Reader, Curtis P/N 1170-001

|         |
|---------|
| J3 - 73 |
|---------|

**3.2.15.2 ELAPSED TIME INDICATOR POWER**

Signal ID: ETI PWR  
 Input/Output: Input  
 Signal Type: 5V Power  
 Function: Elapsed Time Indicator Power  
 Note: \*\*(Bench Use Only)\*\* with ETI Reader, Curtis P/N 1170-001

|        |
|--------|
| J3 -79 |
|--------|

**3.3 Configuration Strapping****Table 5 Transaero EFIS Configuration**

| Signal Name   | EFI pin | PADI | PHSI | CADI | CHSI |
|---------------|---------|------|------|------|------|
| PILOT/COPILOT | J3-27   | OPEN | OPEN | GND  | GND  |
| ADI/HSI       | J3-28   | OPEN | GND  | OPEN | GND  |
| AC0           | J3-2    | GND  | GND  | GND  | GND  |
| AC1           | J3-48   | GND  | GND  | GND  | GND  |
| AC2           | J3-49   | OPEN | OPEN | OPEN | OPEN |
| AC3           | J3-50   | GND  | GND  | GND  | GND  |
| AC4           | J3-29   | OPEN | OPEN | OPEN | OPEN |
| AC5           | J3-30   | OPEN | OPEN | OPEN | OPEN |
| AC6           | J2-7    | OPEN | OPEN | OPEN | OPEN |
| PARITY        | J3-1    | OPEN | GND  | GND  | OPEN |